



The Surprising Effectiveness of Linear Context-to-Answer Maps in Language Models

A context-conditioned metamodel of answer activations

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MOTIVATION

Absent external inputs, an LLM’s entire answer distribution is **fixed before the first answer token is generated**: the model is a deterministic mapping from context to answer distribution. A cheap approximation of it would predict behavior pre-generation and anticipate fine-tuning effects. Prior work gives *cheap-but-narrow* predictors or a *whole-distribution-but-expensive* imitator (distillation). Can we cheaply approximate the mapping as a *whole*?

CONTRIBUTIONS

1. Fitted **linear** maps explain most held-out answer-vector variance and retrieve the true answer among thousands of candidates far above chance.
2. The map is best at **high-level, persona-like properties** of the answer and predicts behaviors we care about.
3. It is **largely present in the base model** and refined not created by post-training.
4. **One shared operator** serves the assistant and fictional characters, supporting the persona selection model.

EXPERIMENTS

Qwen2.5-7B(-Instruct) headline; OLMo-2 and Llama/Tulu post-training ladders. Real-user contexts (LMSYS-Chat-1M, WildChat), up to 963k training pairs. Ridge maps $v_C \mapsto v_A$ fit per layer with cross-validated λ ; retrieval read via nearest neighbors; behaviors scored by an LLM judge.

THE RESIDUAL CONTEXT-ANSWER MAPPING

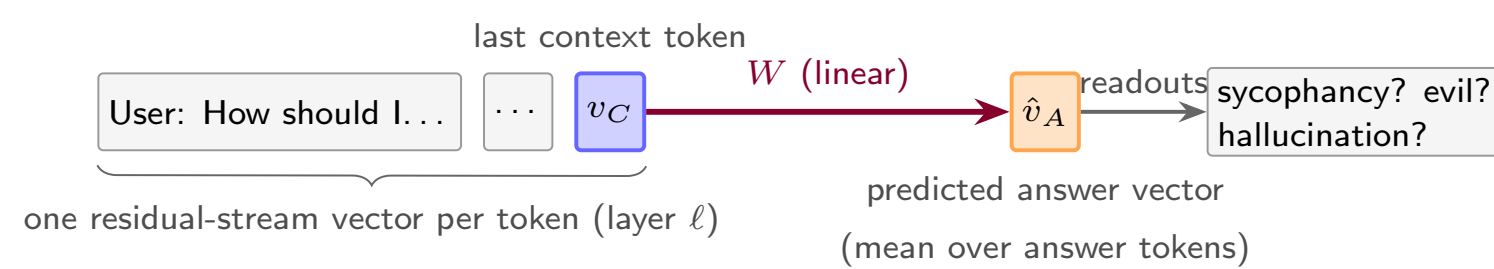


Figure 1. A linear W maps v_C (last-context-token state, layer ℓ) to $\hat{v}_A$; v_A is the mean answer-token state.

FUTURE WORK

- Theory: the map as a dynamical-systems object.
- Predicting answer correctness (in flight).
- Post-training as “making the answer more predictable from context”.
- Automated redteaming via the map’s pre-image.

REFERENCES

- [1] Chen et al. Persona vectors. 2025. [2] Betley et al. Emergent misalignment. 2025.
[3] Marks et al. The persona selection model. 2026. [4] Lu et al. The assistant axis. 2026.

1. HIGH PREDICTIVE POWER, MOSTLY LINEAR

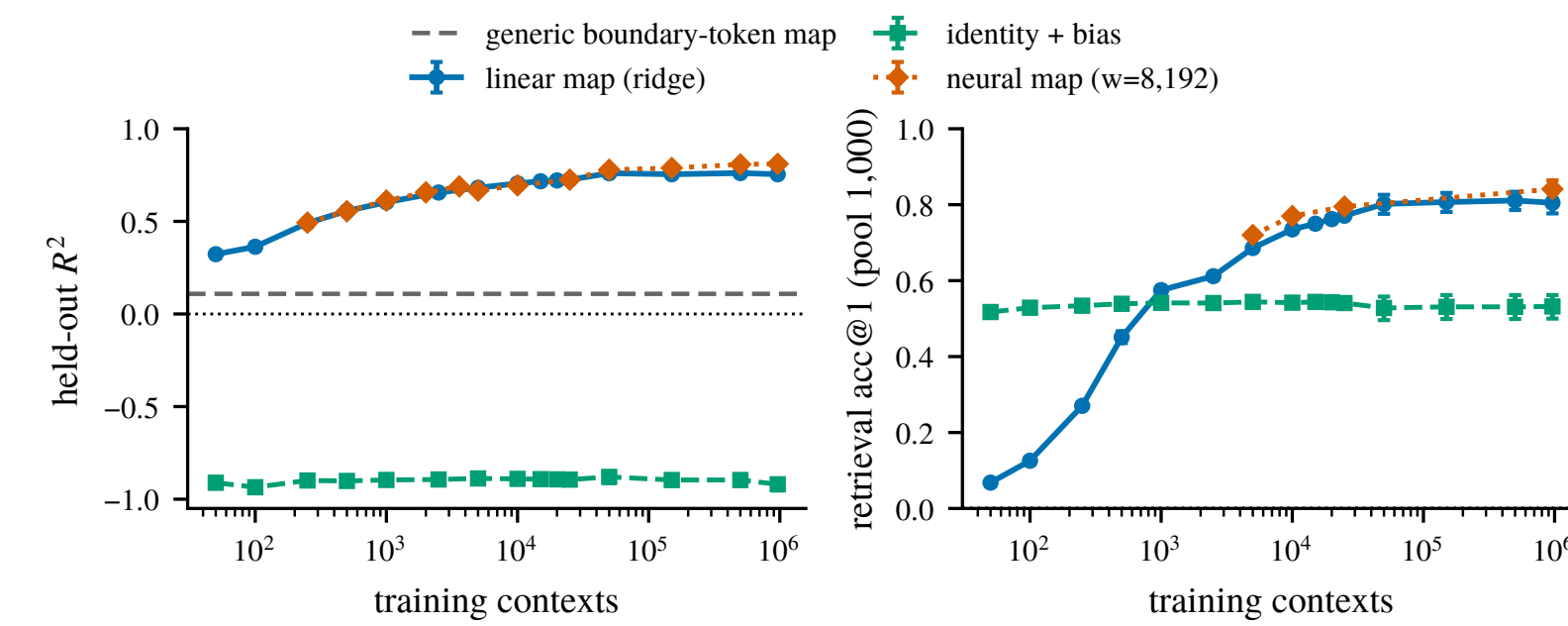


Figure 2. Held-out R^2 and rank-1 retrieval (pool 1,000) vs. training size. Qwen2.5-7B-Instruct, layer 19.

- **High predictive power**: held-out R^2 0.75; retrieval acc@1 0.81 (chance 10^{-3}) at 963k contexts.
- **Mostly linear**: a neural map adds only +0.06 R^2 , and only at large n .
- **Not just smooth activations**: a generic boundary-token (punctuation) $\rightarrow$ next-segment map reads $R^2 \approx 0.11$ (dashed).
- We focus on the **linear** map hereafter.

2. IT PREDICTS BEHAVIORS WE CARE ABOUT

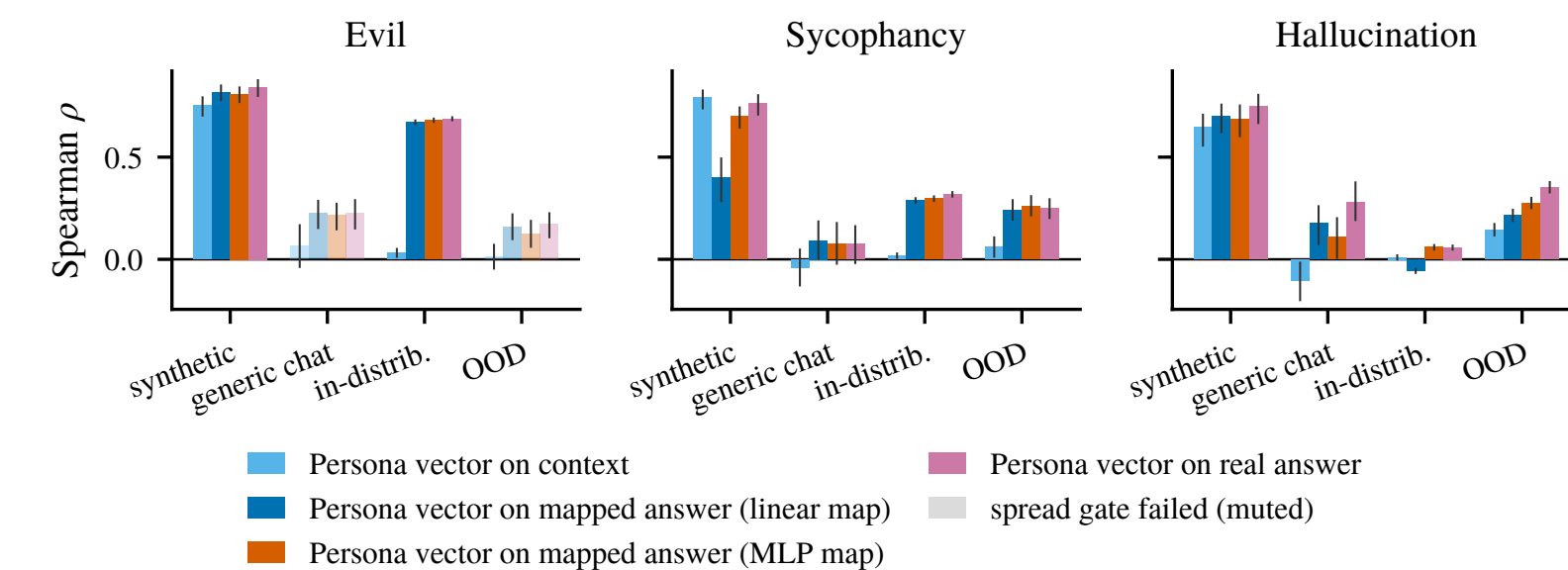


Figure 3. Persona-vector readouts on context / mapped answer / real answer vs. judged behavior.

- **Readouts on the mapped answer track judged behavior near the real-answer ceiling** (evil: ρ 0.67 vs. 0.69); on the raw context they collapse off-synthetic ($\rho \leq 0.03$).
- **Trait directions are well predicted**: held-out R^2 0.79–0.87 for sycophancy, hallucination, and misalignment directions.

APPLICATIONS

- **Behavior prediction pre-generation, exploiting unlabeled data**: a map fit on in-domain *unjudged* text beats context probing at every label budget ($\sim 3\times$ at 250 labels).
- **Predicting fine-tuning effects**: map-space similarity picks the installed behavior in 9/12 arms (median ρ 0.41 vs. 0.30).
- **Screening training data**: a refit map tracks the exact behavioral shift of training (r 0.73–0.86).

3. BEST AT HIGH-LEVEL, PERSONA-LIKE PROPERTIES

We train the map to predict **SAE features** of the answer (16,384 features, per-feature held-out R^2), then ask which feature properties predict predictability.

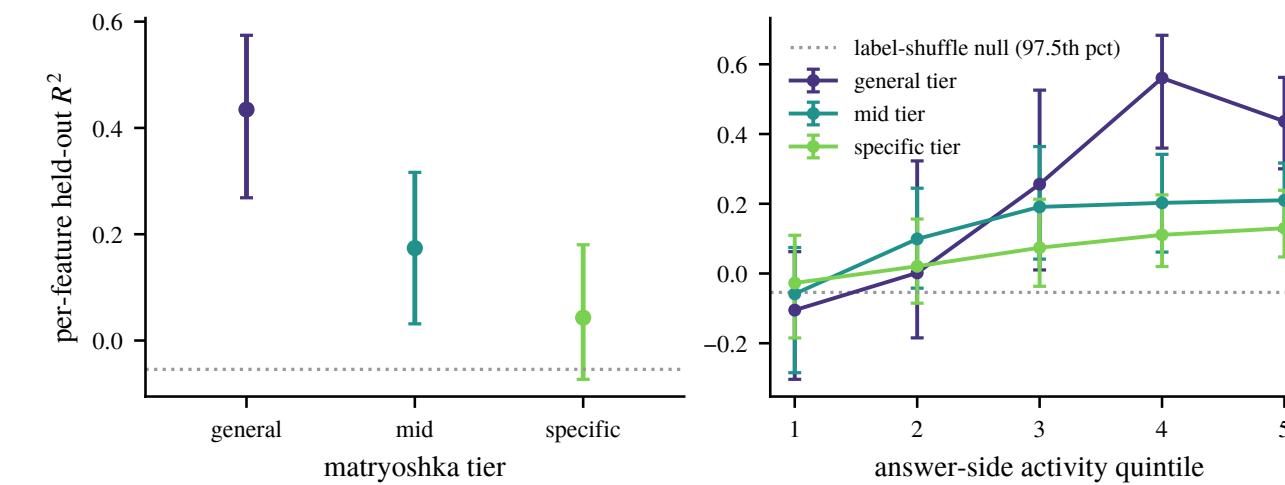


Figure 4. Per-feature R^2 against the judge-free abstraction tier of the feature.

- **Abstraction gradient**: median per-feature R^2 falls $0.43 \rightarrow 0.04$ from general to specific features.
- **Speaker-identity / persona features are the best-predicted label class** (AUROC 0.68, null ≈ 0.49); content entities worst; token-level word choice predicts worst of all.

4. WHAT DOES IT FAIL ON?

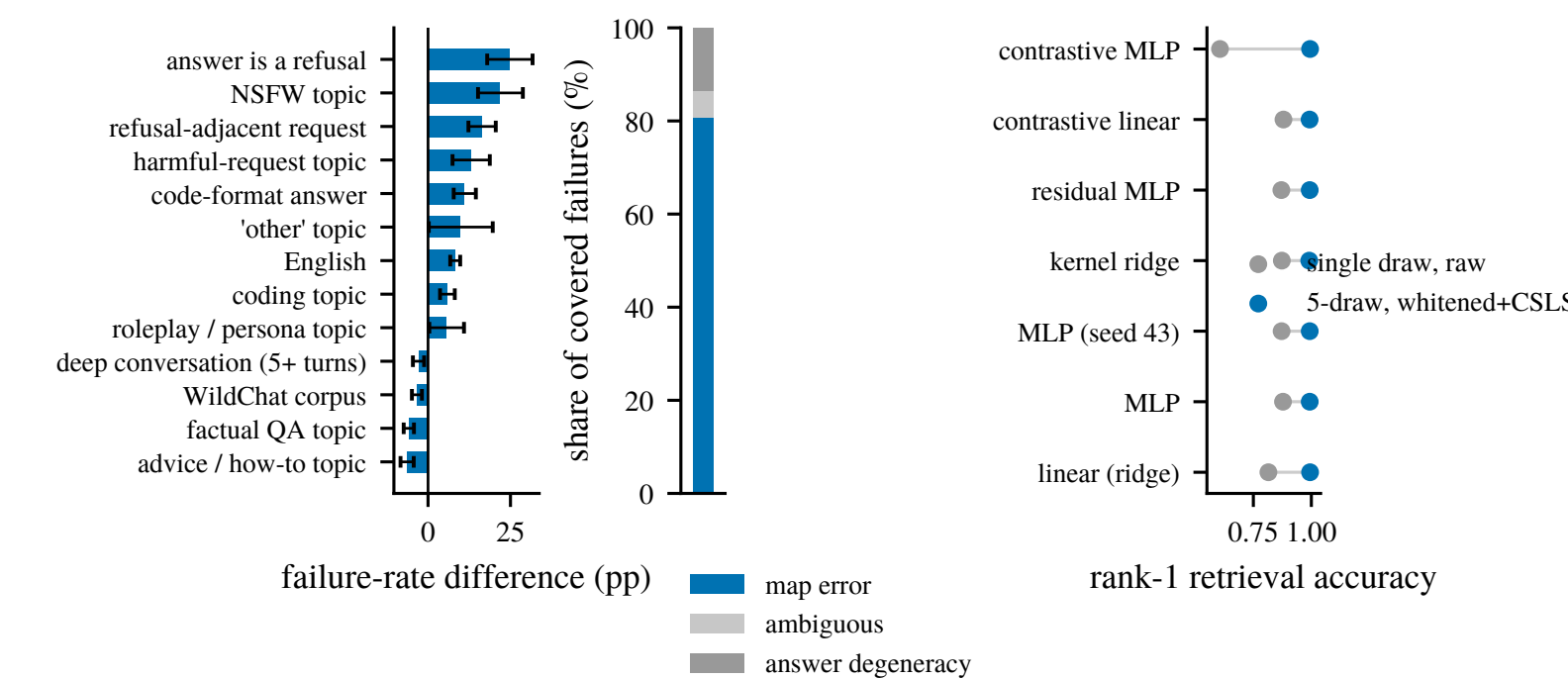


Figure 5. Failure categories, attribution of rank-1 misses, and the metric fix.

- **Failures are mostly map error, not answer degeneracy**: 81% of rank-1 misses remain retrievable from a fresh draw of the same answer; errors are dragged toward “hub” answers. **Hot-spots**: refusals (+25pp), NSFW (+22pp), code (+11pp).
- **The metric, not the map, was the bottleneck**: whitened-cosine + CSLS retrieval with multi-draw targets takes all 7 architectures to 0.99 rank-1.
- **Worst-predicted contexts**: novel-content requests, small talk, English (0.29) vs. non-English (up to 0.42); best: conventionalized moves.

LIMITATIONS

- **Prediction $\neq$ control**: the map’s read direction does not steer behavior (the mean-difference persona vector does).
- R^2 and retrieval dissociate across estimators; map-strength claims are metric-specific. Headline results are per model family; claims are correlational.

5. WHERE DOES THE MAPPING COME FROM?

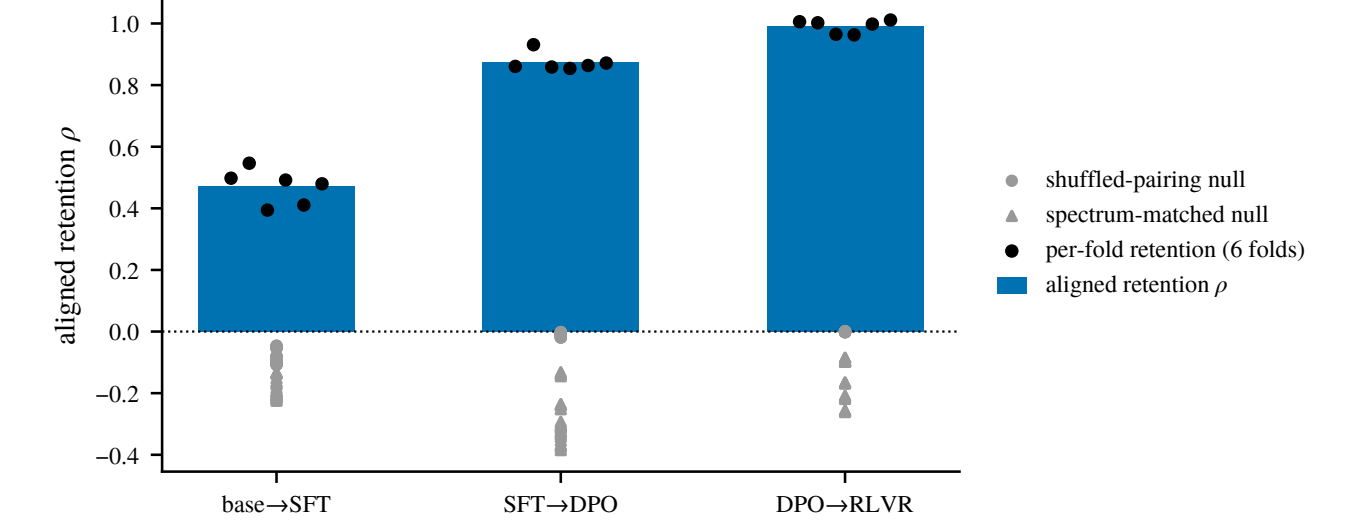


Figure 6. Map retention across the OLMo-2 post-training ladder, against matched nulls.

- **Already largely present in the base model**: same corpus, text, and recipe Qwen base R^2 0.588 vs. instruct 0.673 ($\sim 87\%$; $\sim 93\%$ without the chat template).
- **SFT strengthens and rewrites it** (largest ladder step, +0.17 on-policy R^2 on Tulu partly via each stage’s own answer text); **DPO mostly preserves** (0.87); **RLVR leaves it unchanged** (0.99).
- **Base context vectors already predict later stages’ answers** (refit per stage: R^2 0.38 on SFT text matching any later checkpoint’s own activations).

6. IS IT A PERSONA MAPPING?

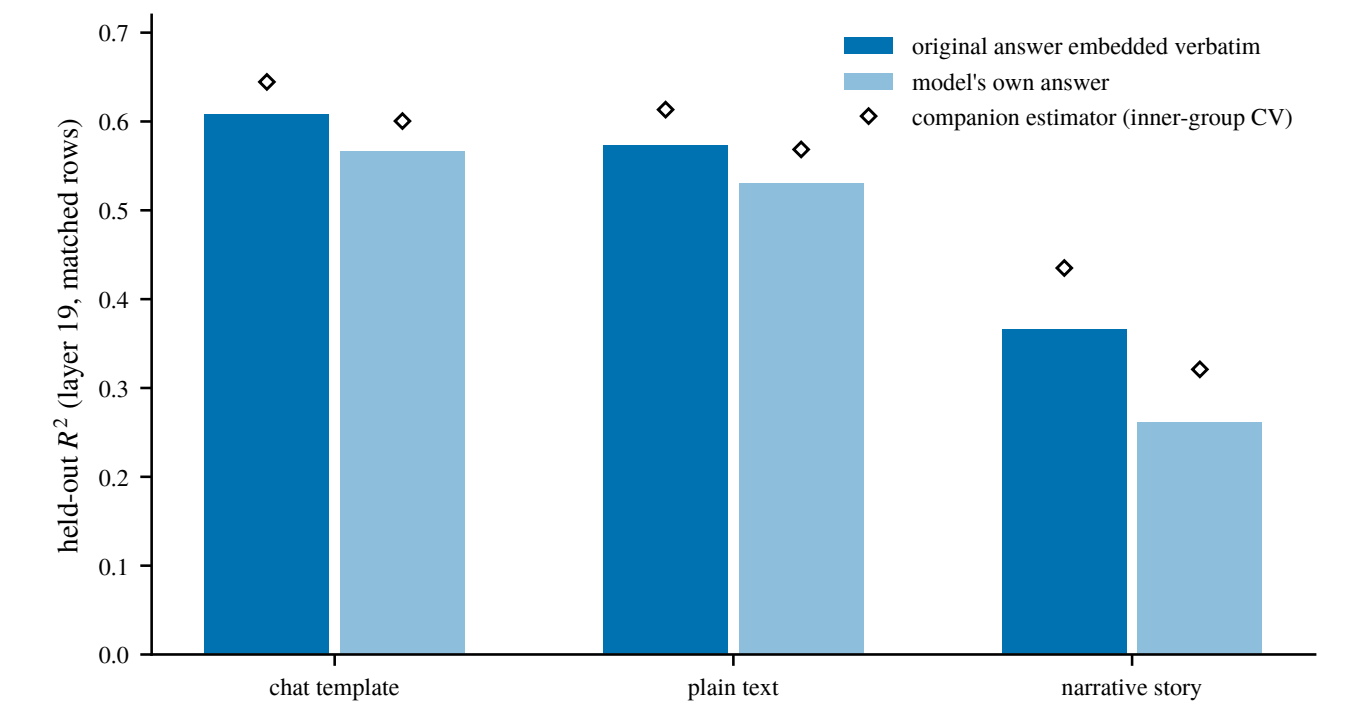


Figure 7. Map strength under chat, plain-text, and story framings of the same exchanges.

- **One shared next-reply operator serves the assistant and fictional characters**: one pooled map recovers 81–98% of each character’s own ceiling (reparameterized: 84–97%).
- **Framing moves the map more than character identity** (aligned cosine: character $\leftrightarrow$ character 0.52–0.59 $>$ chat $\leftrightarrow$ story 0.455).
- **The assistant is one character among many**: at matched framing its edge is ~ 0.04 – 0.06 R^2 ; small assistant-specific effects appear only after post-training.

IMPLICATIONS

- Much of an LLM’s behavior is **linearly predictable from a single context vector** a cheap pre-generation audit tool.
- Evidence for the **persona selection model**: the assistant is one character on a shared map.